

Aishwarya Akula

aishwaryaakula754@gmail.com | +91 7842534034 | Hyderabad, India

OBJECTIVE

B. Tech graduate in Electronics and Communication Engineering with a strong interest in Embedded Systems and real-time applications. Possess a solid foundation in electronics, and system design, with hands-on experience through academic projects. Skilled in Embedded C, basic Python, and hardware interfacing, with a good understanding of communication protocols and embedded development concepts. Seeking a full-time role in Embedded Systems where I can apply my knowledge of hardware-software integration, debugging, and system optimization to develop reliable and efficient embedded solutions, while continuously learning and growing in the field.

PROJECTS

Empowering Libraries Using IoT:

Automated Book Sorting System: Designed an intelligent system that automatically sorts books based on color using Arduino and color sensing technology, reducing manual effort in libraries. Sensor & Control Mechanism, Integrated color sensors to detect book categories, with a microcontroller processing inputs and controlling motors to direct books into appropriate sections. Improved Accessibility & Efficiency, Enhanced library management by enabling faster book identification and arrangement, improving user experience and operational efficiency.

3D Scanner for Measuring Objects:

Automated Measurement System: Designed a cost-effective 3D scanning system using IR sensors to measure object dimensions accurately. Sensor & Data Processing, Implemented automated data acquisition and processing techniques to enhance precision under varying environmental conditions. Industrial Application, developed a scalable solution suitable for manufacturing, quality control, and logistics, improving efficiency and measurement accuracy.

Designed a 16-bit Vedic Multiplier Using Modified Square Root CSA Adder:

High-Speed Multiplier Design: Implemented a 16-bit Vedic multiplier architecture for faster arithmetic computations in digital systems. Optimized Adder Integration, integrated a Modified Square Root Carry Select Adder (CSA) to reduce propagation delay and enhance performance. Performance Improvement, achieved higher computational efficiency and speed compared to conventional multiplier designs, making it suitable for high-speed applications.

SKILLS

Programming Languages: Embedded C, Python, Basic, C Programming

Platforms/ Tools: Arduino IDE, SPICE (for circuit simulation), Tinkercad

Communication Protocols: UART, I2C, SPI

Hardware Design & Prototyping: Circuit Simulation, Microcontrollers

EDUCATION

Hyderabad Institute of Technology and Management (HITAM), Hyderabad

2021 – 2025

B. Tech- Electronics and Communication Engineering CGPA: 7.83

Sri Chaitanya Junior College, Hyderabad

2019-2021

Intermediate GPA: 8.52

Montessori High School, Hyderabad

2019

GPA: 8.8

CERTIFICATES

python certificate from Innomatics Research Labs

Embedded Systems Programming (Arduino / 8051) -Udemy

Introduction to IoT- Coursera